Perfect Competition and Monopoly

C211: Global Economics for Managers

Western Governors University

PERFECT COMPETITION

The History of Perfect Competition

First developed in the 1860s

Developed as a representation of perfectly efficient capitalism

Developed at the same time as the Monopoly model

-which as meant to describe broken capitalism

The History of Perfect Competition

Developed during industrial revolution capitalism

During time when most business were owner operated

Most firms produced a single good in a factory or farmland

Charles Dickens books are about this time

Perfect Competition

Basics:

Identical good

-No difference between firms

No barriers to entry

-Cheap and easy to enter

Cannot affect market price

-Individual firms can't change the market price when they change quantity

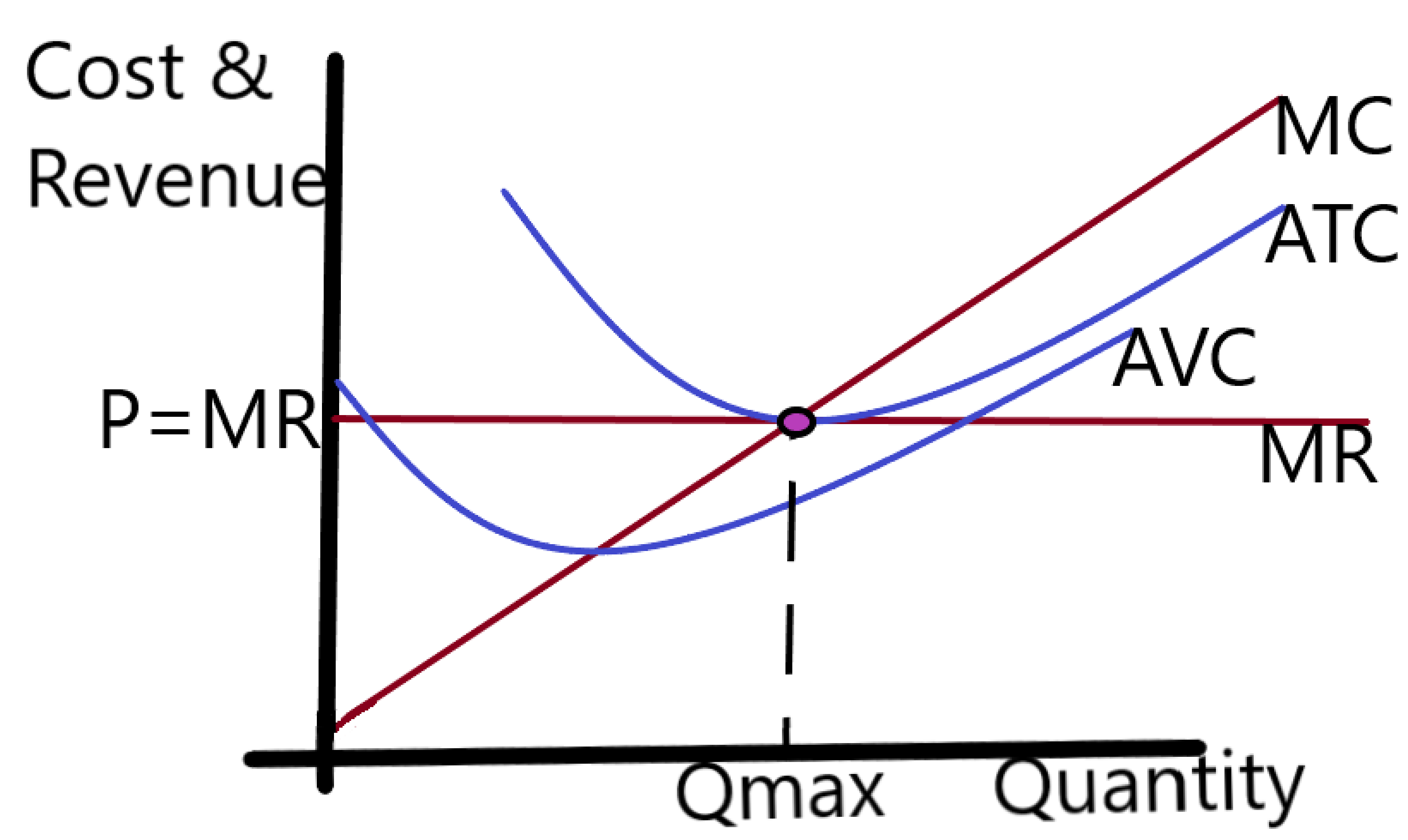
P=MR

P=MC (at profit max)

Examples:

-Chicken Farmer, Egg Producer, Wheat Farmer, Generics

Long-Run Graph



Terms

ATC ➔ Average Total Cost

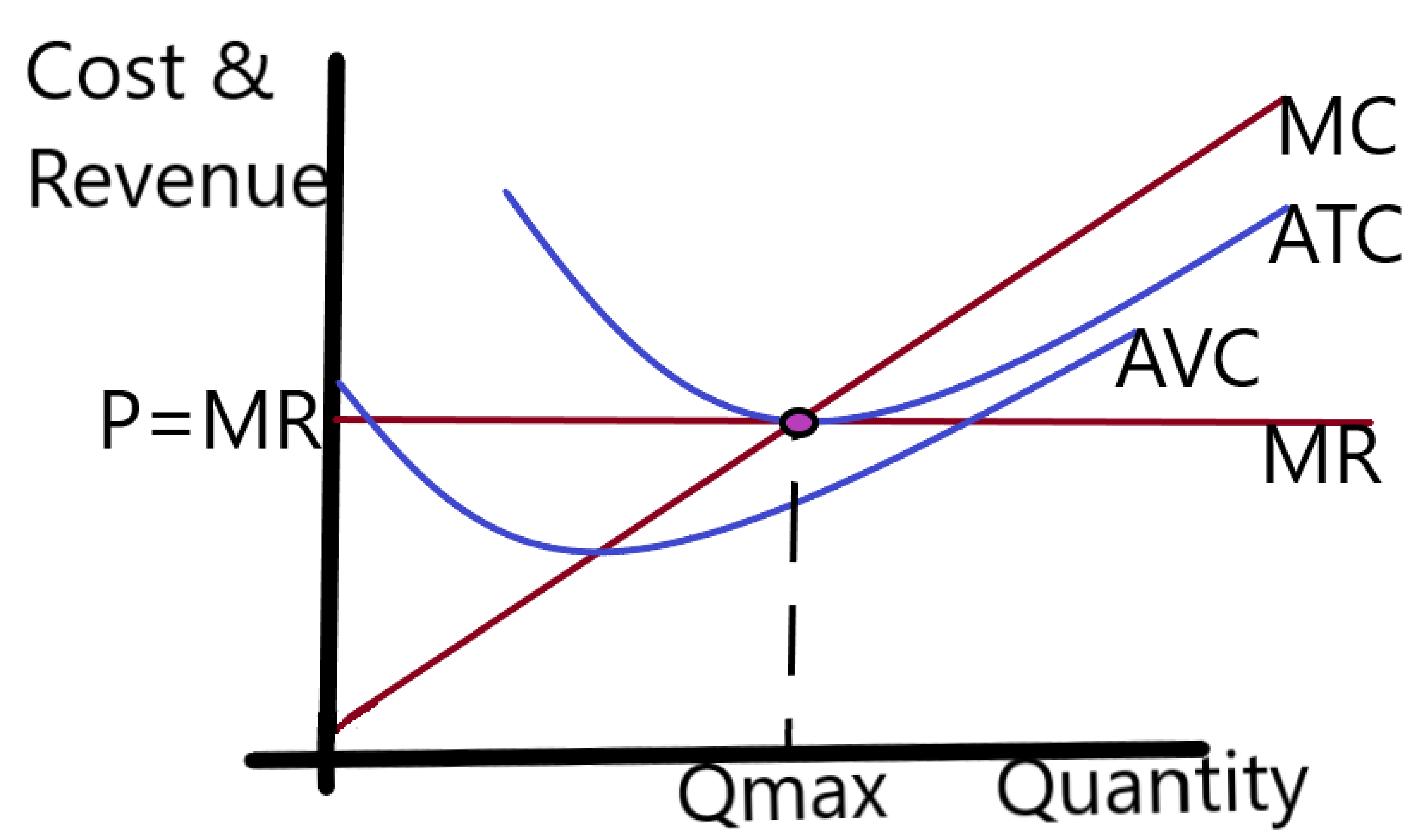
AVC➔ Average Variable Cost

MC➔ Marginal Cost

MR➔ Marginal Revenue

Qmax➔ profit maximizing Quantity

Long-Run



Price given by market and can't be changed by individual firm

P=MR ➔ No tradeoff

Qmax is efficient scale (minimum of ATC)

P=ATC ➔ zero economic profit

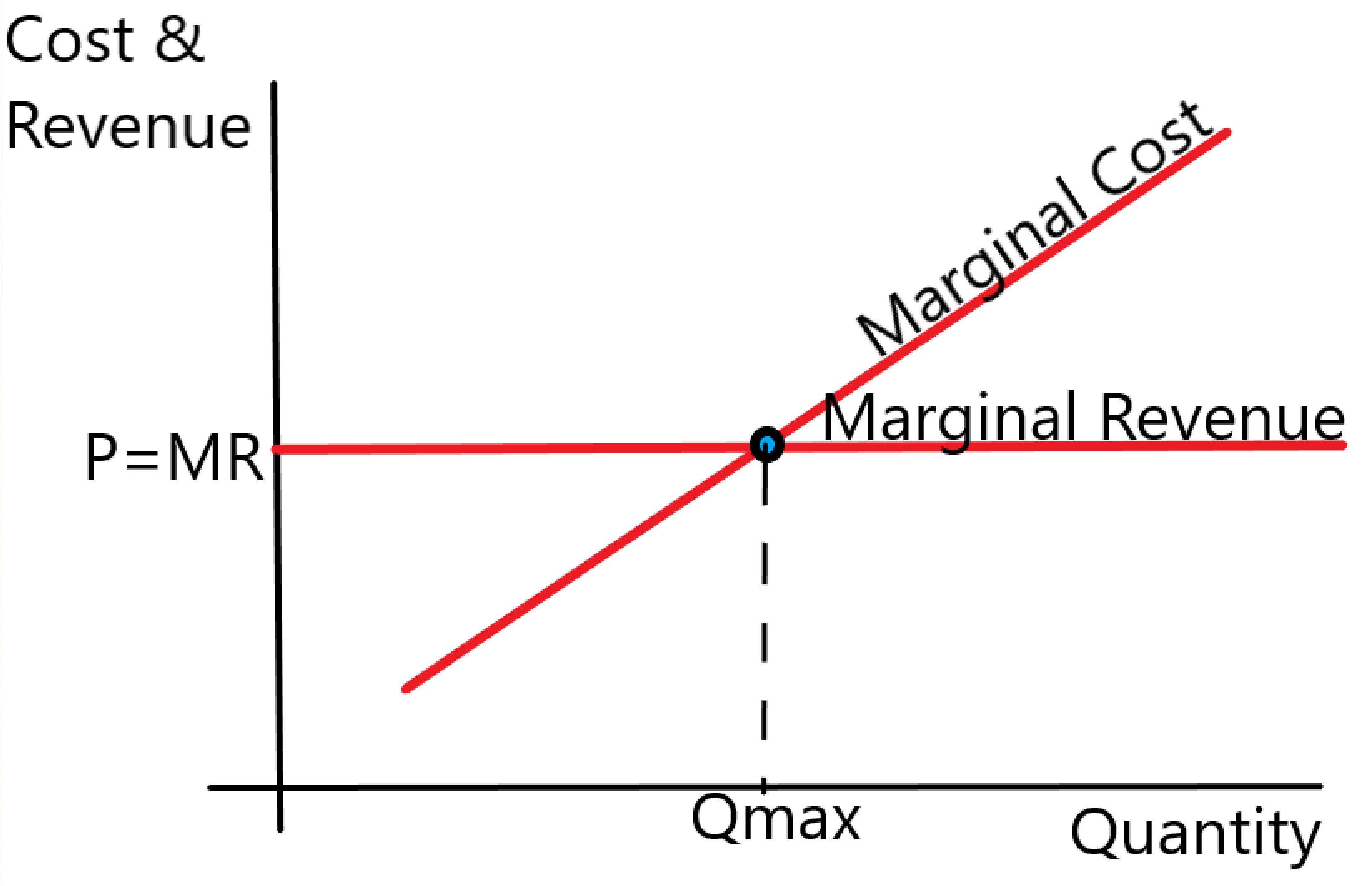
Similar graph on Ch14-2b

Profit Maximization

-Always occurs where **Marginal Cost Equals Marginal Revenue**

-This is true for ALL market types

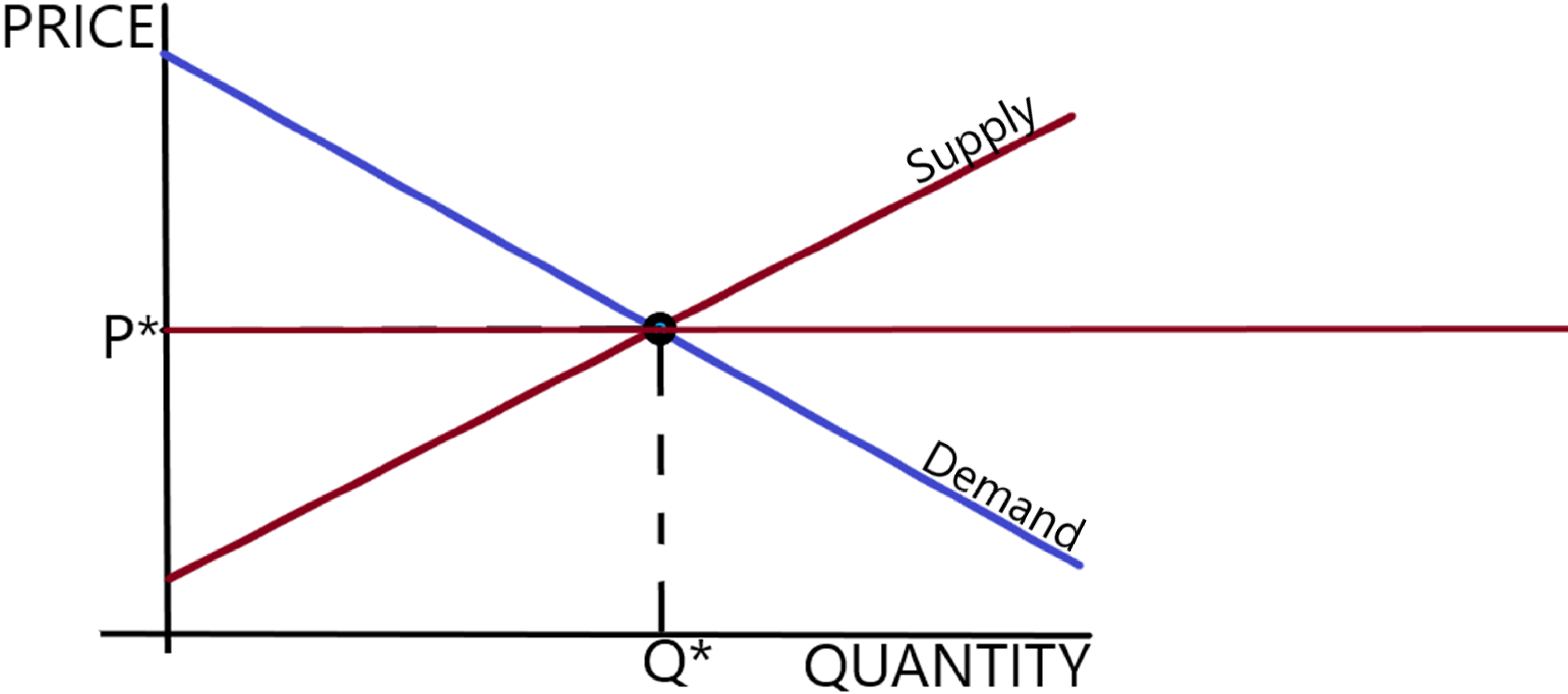
See Chapter 14-2B



-Because for Perfect Competition P=MR we can also say P=MC at profit max (see ch16-2c)

Price-Taker

The price is given by the market.



Price-Taker

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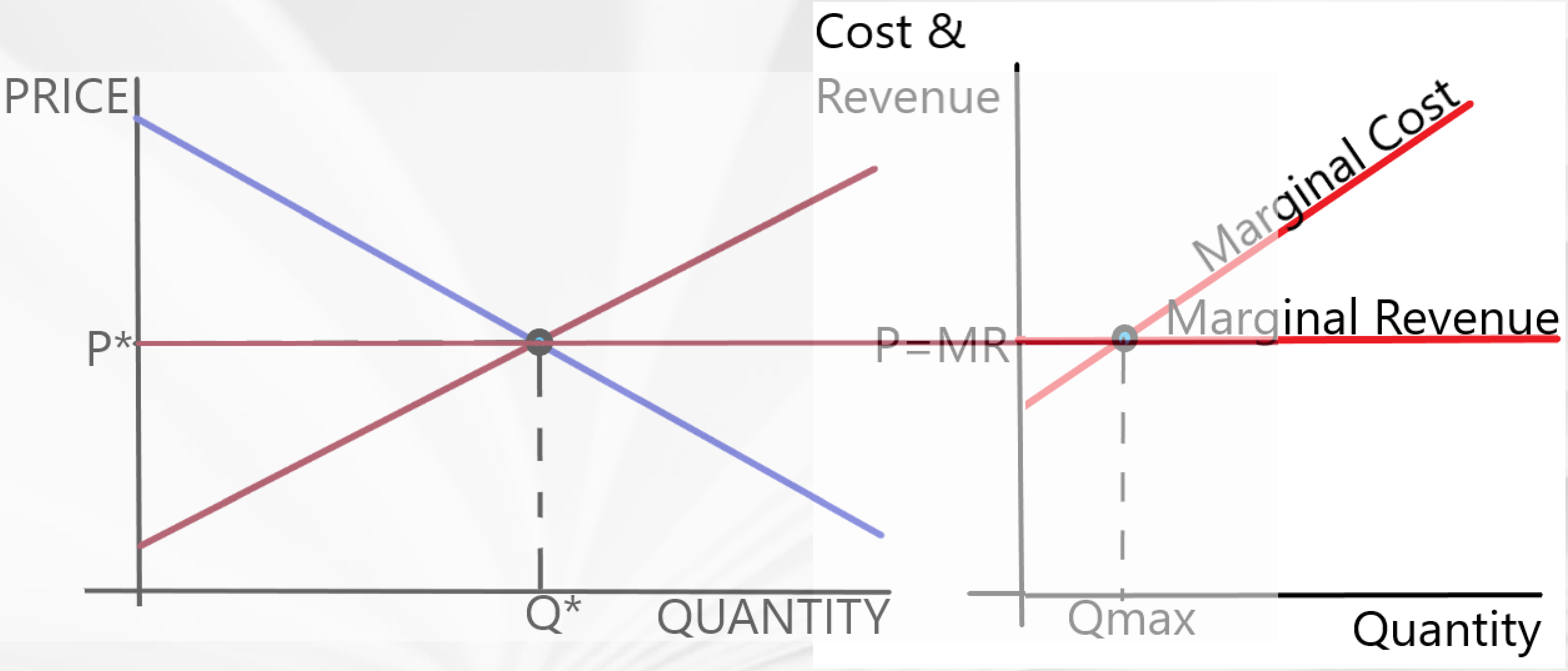
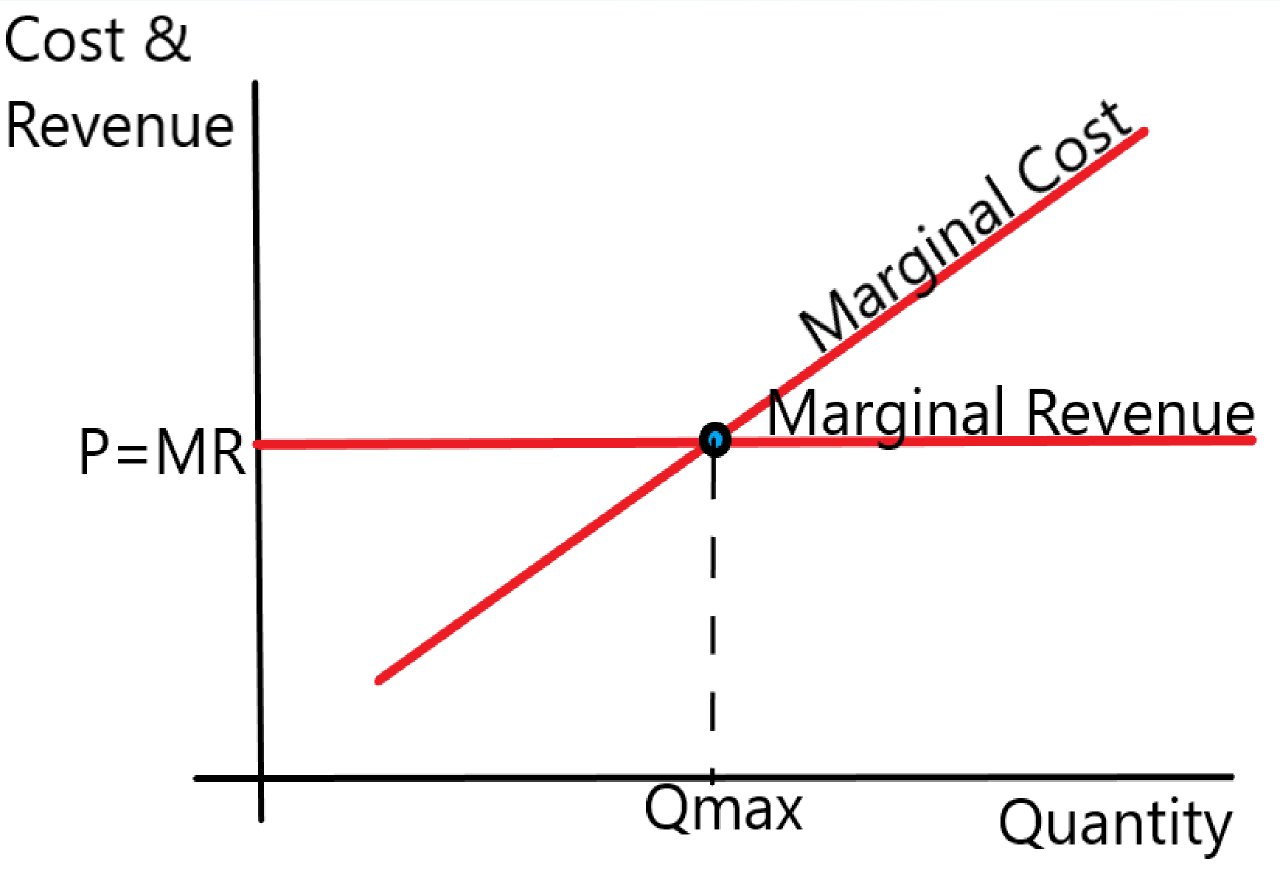


Why does Price equal Marginal Revenue?



Marginal Revenue is the change in total Revenue when a single unit is sold

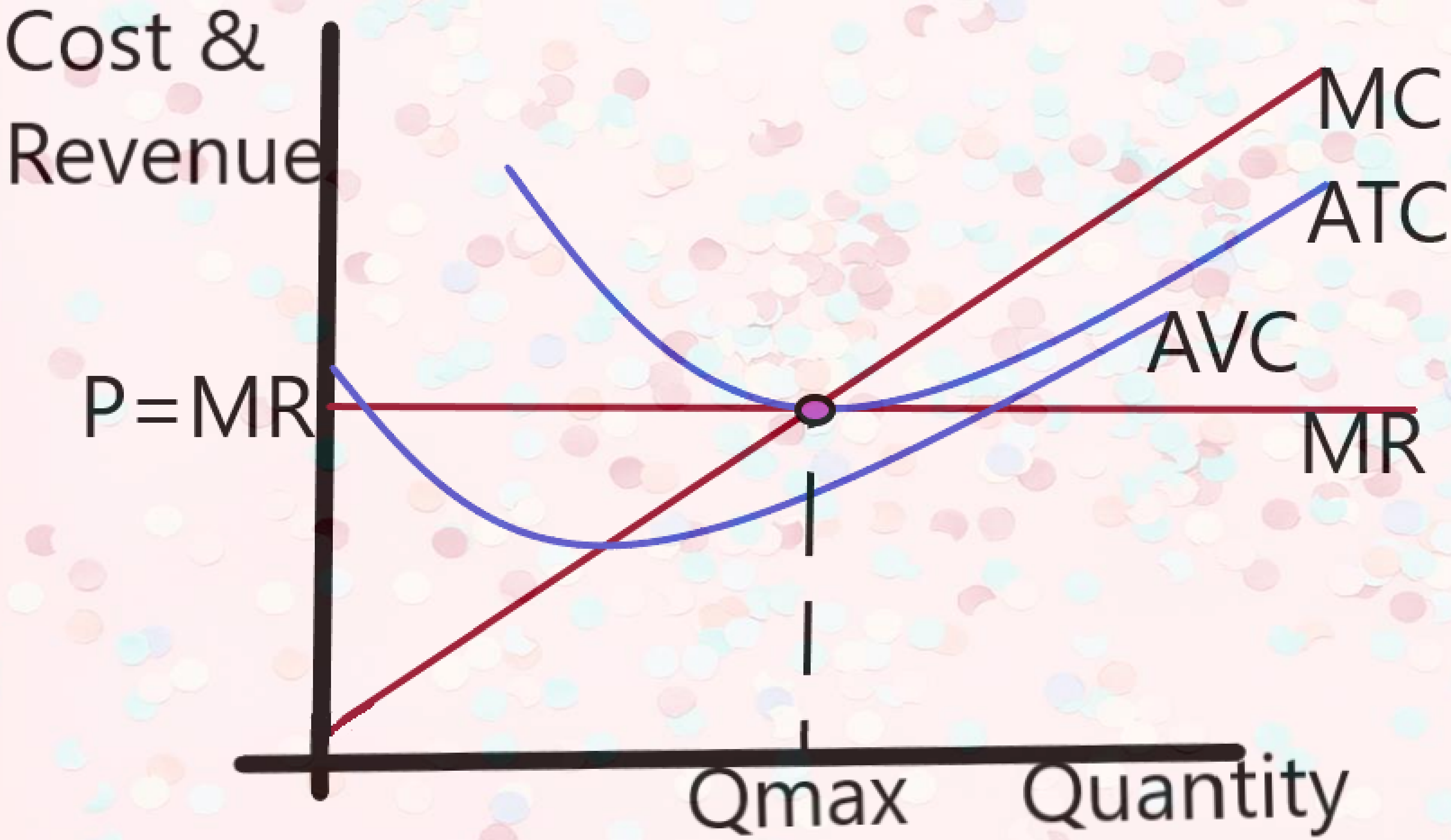
When the Competitive firm produces another unit, it doesn't affect price

As a result, the way Total Revenue changes equals the unchanging Price

Therefore: **P=MR**

Can produce any quantity at given price

Long-Run



Price given by market and can't be changed by individual firm

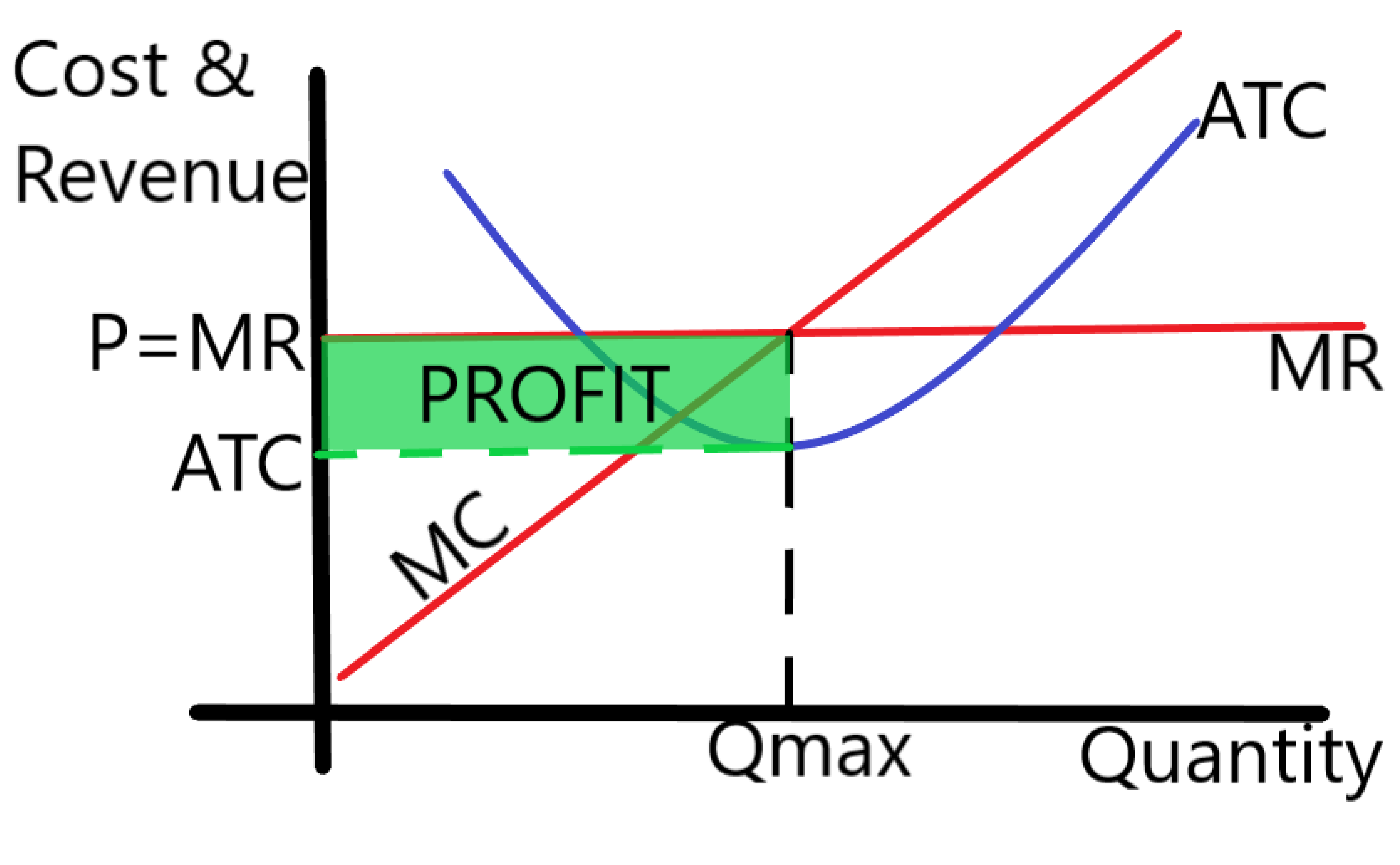
P=MR ➔ No tradeoff

Qmax is efficient scale (minimum of ATC)

P=ATC ➔ zero economic profit

Similar graph on Ch14-2b

Short-Run (2 Possibilities)



Price given by the market

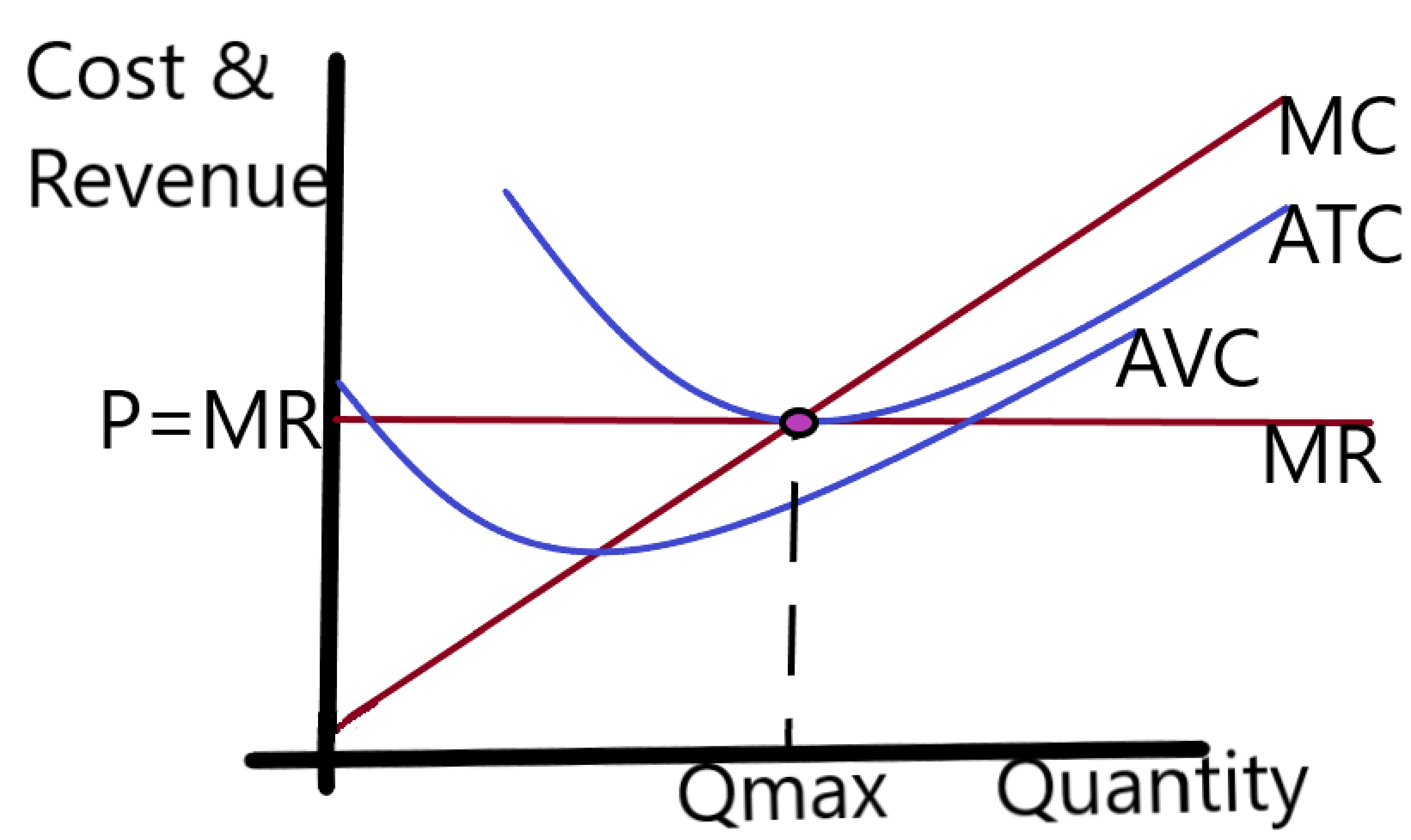
P=MR→No tradeoff

Qmax is where Marginal Revenue Equals Marginal Cost

P>ATC so firm earns above zero economic profits

New firms will enter market

Long-Run



P=ATC ➔ zero economic profit

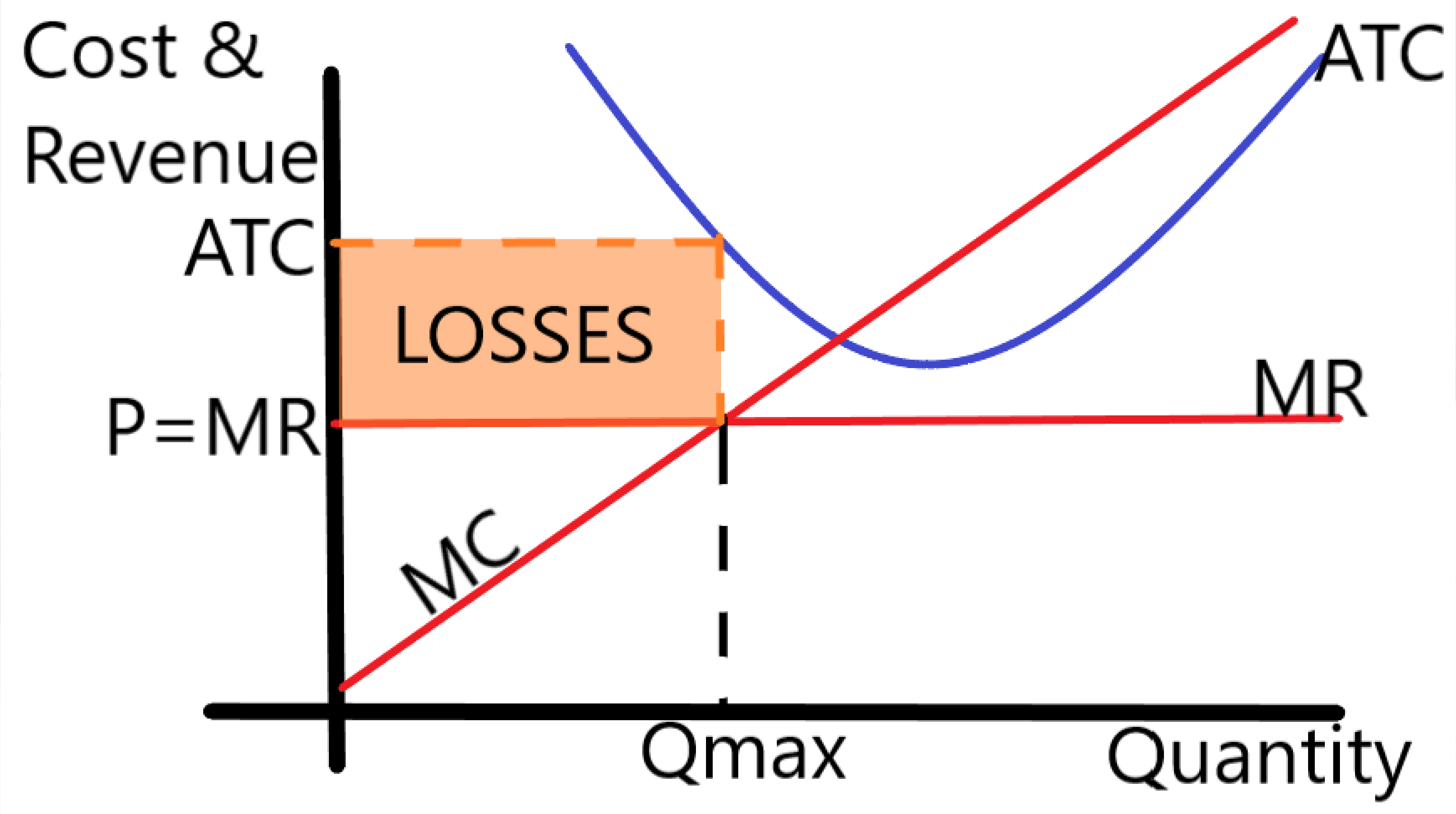
As firms enter quantity in the market rises

As quantity rises price falls

Long-run reasserts itself

Similar graph on Ch14-3d

Short-Run (2 Possibilities)



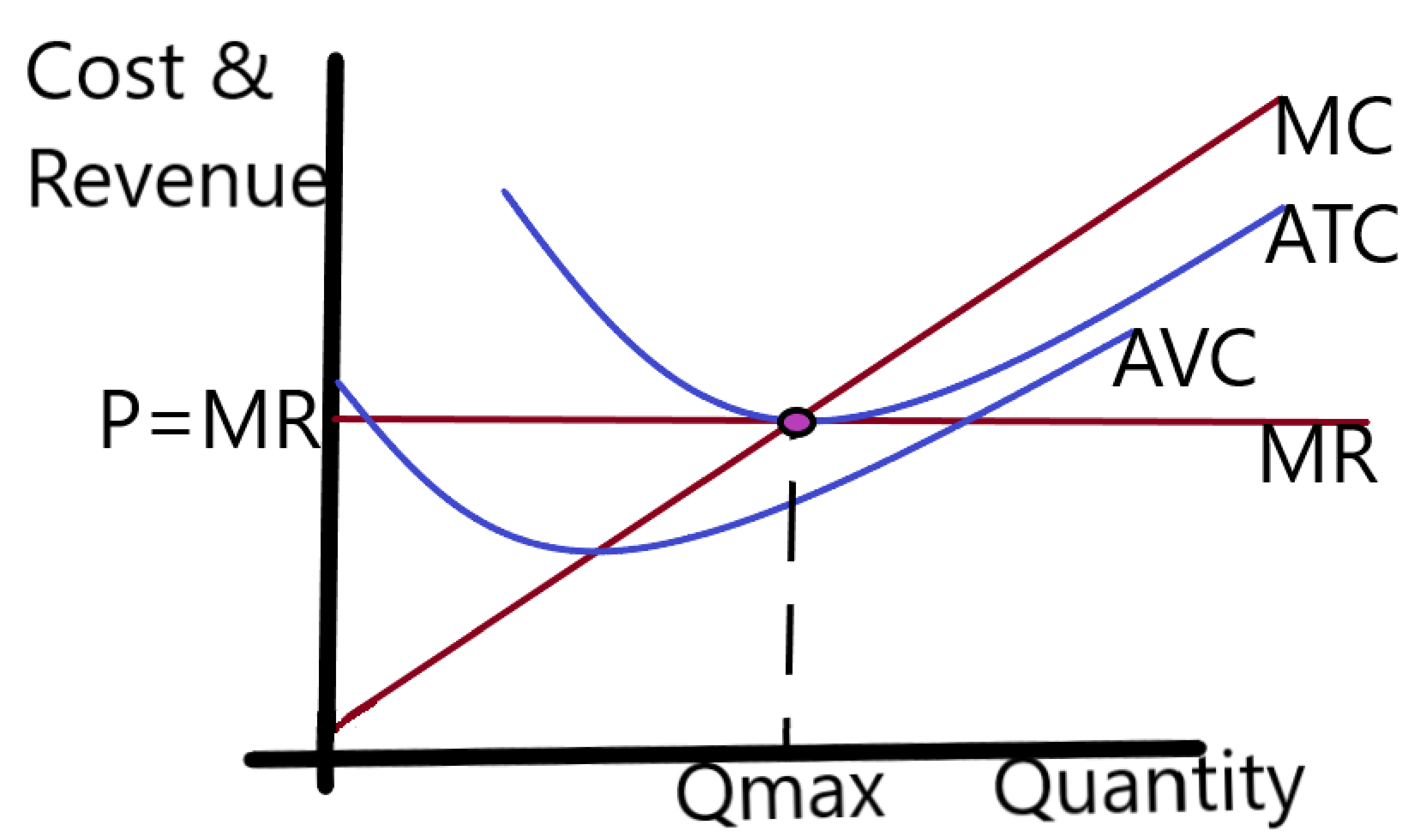
Price given by the market

Qmax is where Marginal Revenue Equals Marginal Cost

P<ATC so firm earns losses

Some firms will shutdown, some exit

Long-Run



P=ATC ➔ zero economic profit

As firms exit quantity in the market falls

As quantity rises price falls

Long-run reasserts itself

Similar graph on Ch14-3d

Perfect Competition

Short-Run Shutdown Rule

(see ch14-2c)

Short-Run→ Defined as having at least one input which is fixed

Like the building of a factory, or a lease contract

These are "Fixed Costs" and do not vary with quantity produced

Perfect Competition

Short-Run Shutdown Rule

(see ch14-2c)

Fixed Cost cannot be changed no matter the quantity produced

Variable Costs→ those costs that come directly from production

These costs are zero if nothing is produced

Ex: Cups and coffee for a coffee shop, Oil and metal for a car factory, wages

Perfect Competition

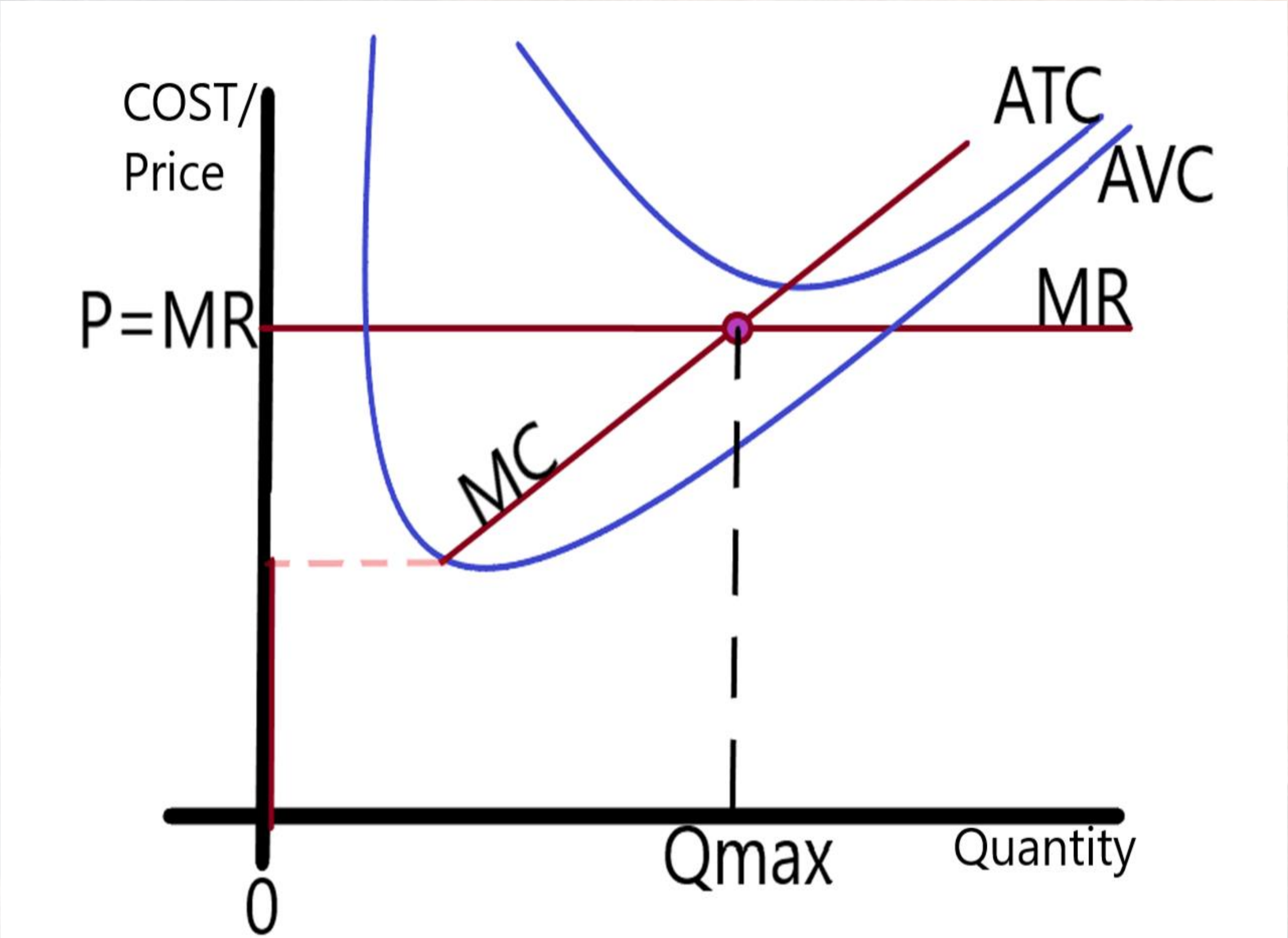
Short-Run Shutdown Rule (see ch14-2c)

Fixed Cost cannot be changed no matter the quantity produced

Because our choice of quantity cannot affect fixed costs they are a “Sunk Cost”

So we do not focus on them in the Short-Run

Perfect Competition



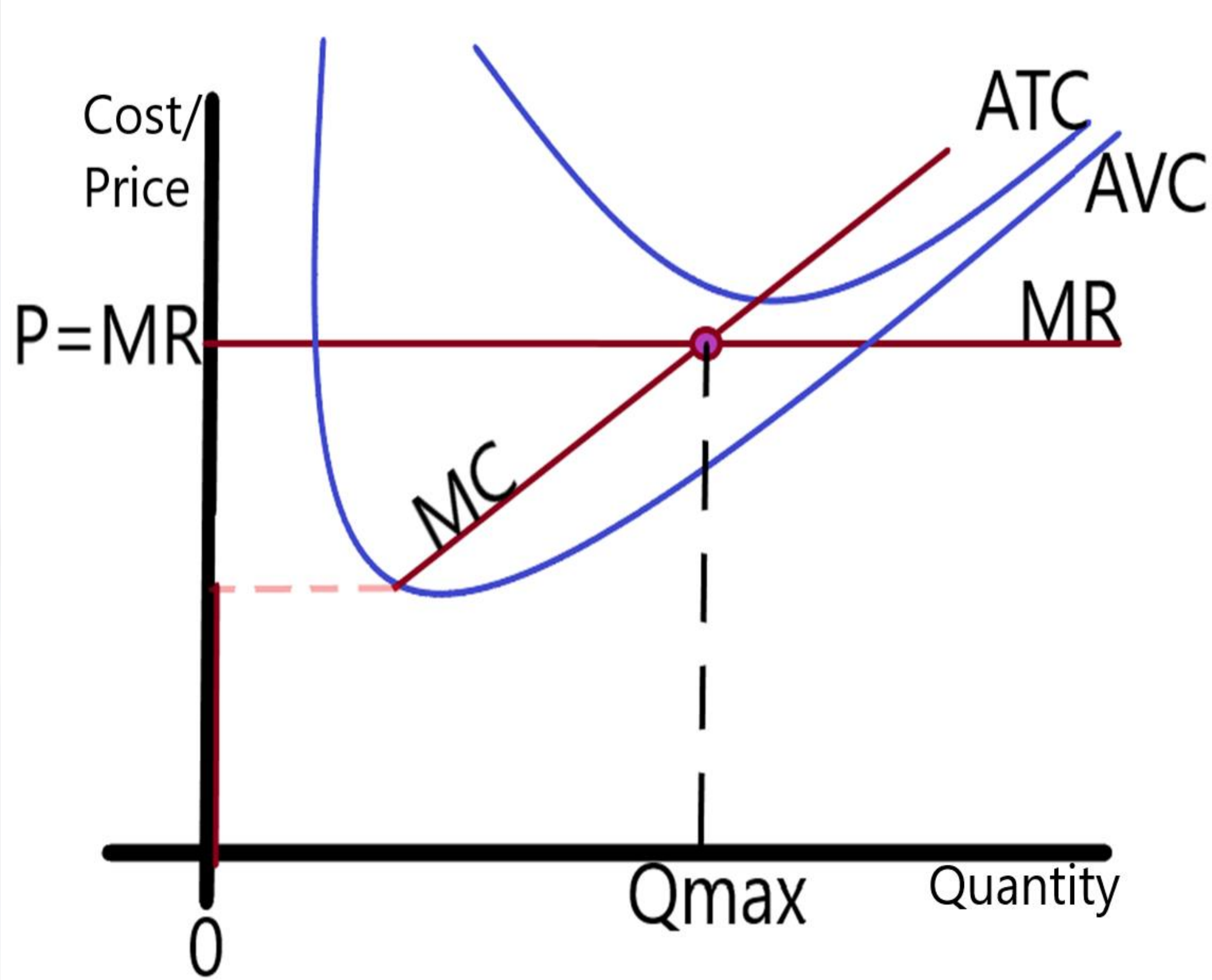
Short-Run Shutdown Rule (see ch14-2c)

Fixed Cost cannot be changed in the Short-Run

So instead focus on Variable Costs only

If revenue is less than those Variable Costs we do not produce anything and those costs go to zero.

Perfect Competition



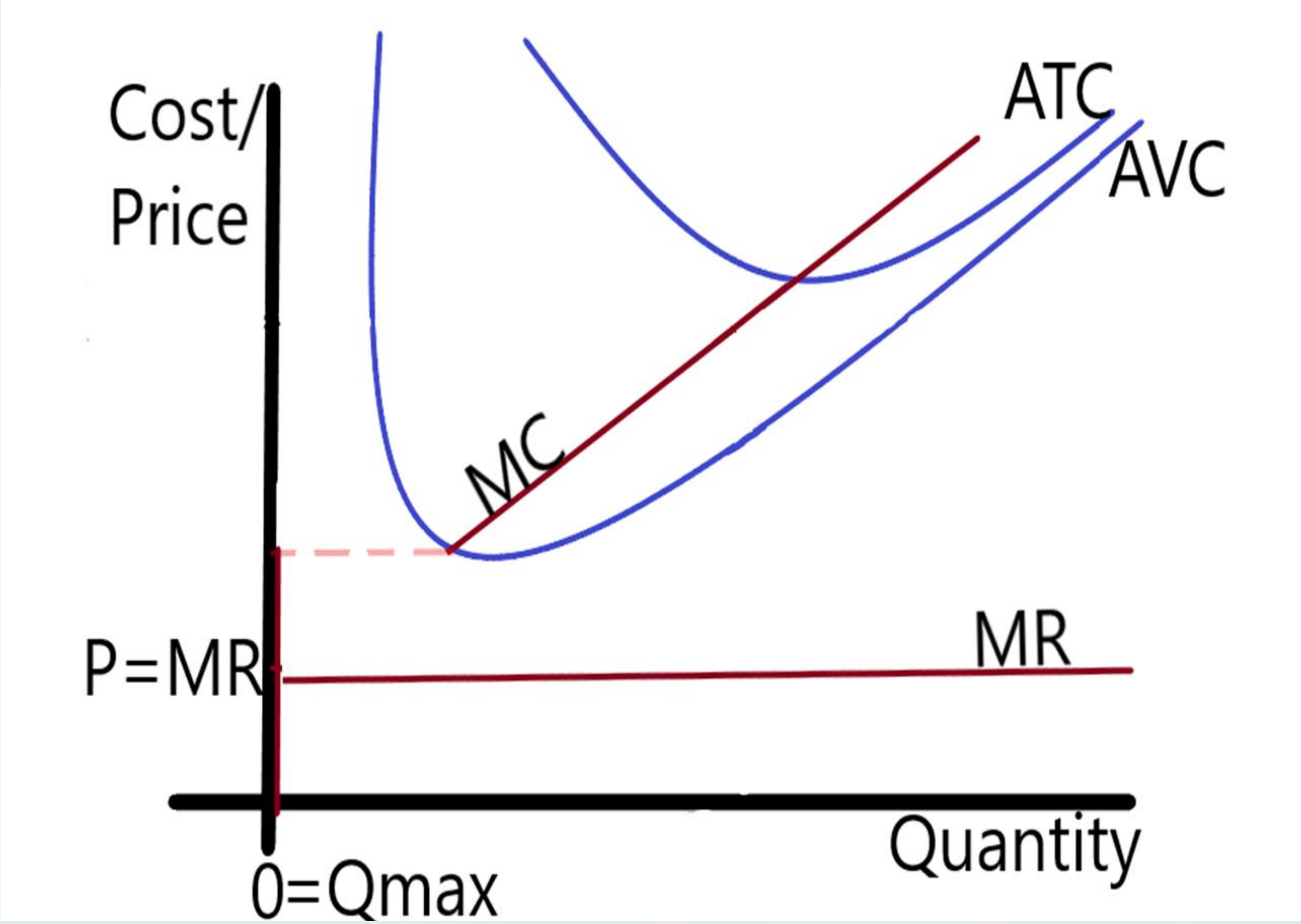
Short-Run Shutdown Rule (see ch14-2c)

The Supply curve for the firm IS the Marginal Cost curve

If price is above Average Variable Cost we produce (pictured)

If revenue is less than those Variable Costs we do not produce anything and those costs go to zero.

Perfect Competition



Short-Run Shutdown Rule (see ch14-2c)

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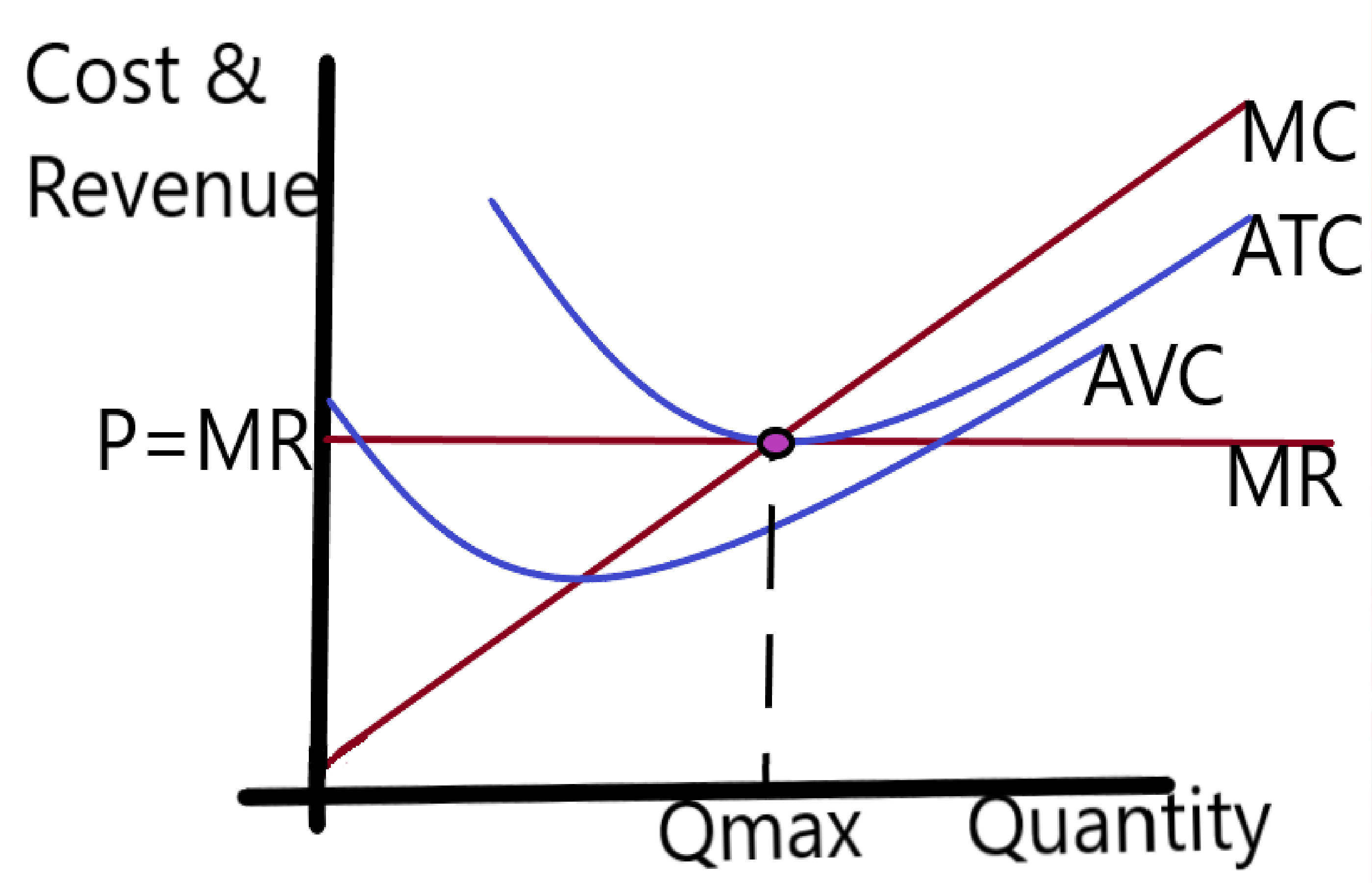
Perfect Competition



Long-Run EXIT

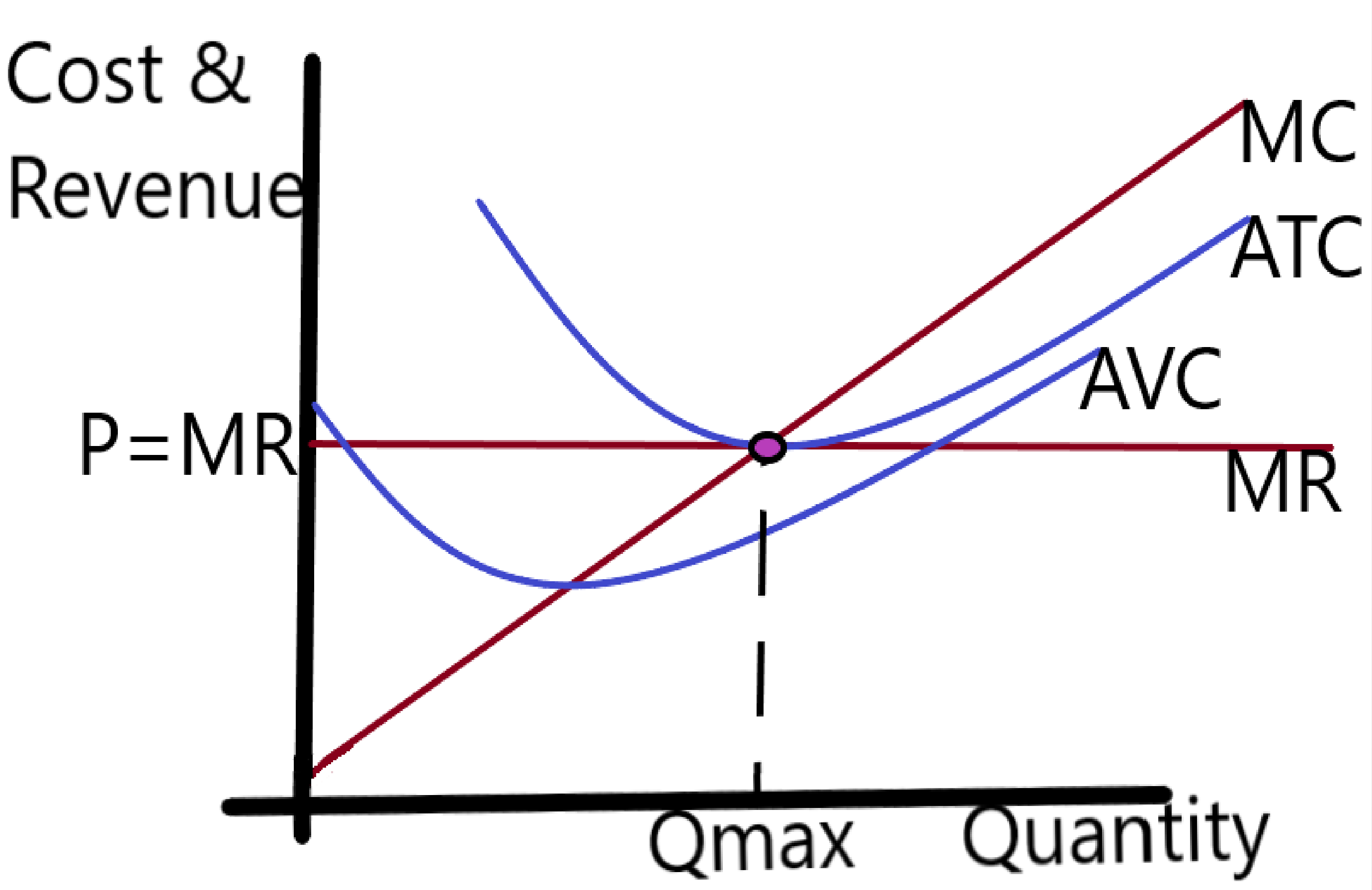
In the long-run Fixed costs are variable again

Both Fixed and Variable Costs must be considered

We care if our Revenue will cover Total Cost

P<ATC➔ Exit

Perfect Competition



Long-Run EXIT

In the long-run Fixed costs are variable again

Both Fixed and Variable Costs must be considered

We care if our Total Revenue will cover Total Cost

P<ATC➔ Exit

TR<TC➔ Exit

Monopoly

Basics:

Unique good

-one firm, one good

Impossible barriers to entry Ch15-1

-No competition

Can affect market price

-Firm's changing quantity can change the market price

-Price determined by Demand Curve

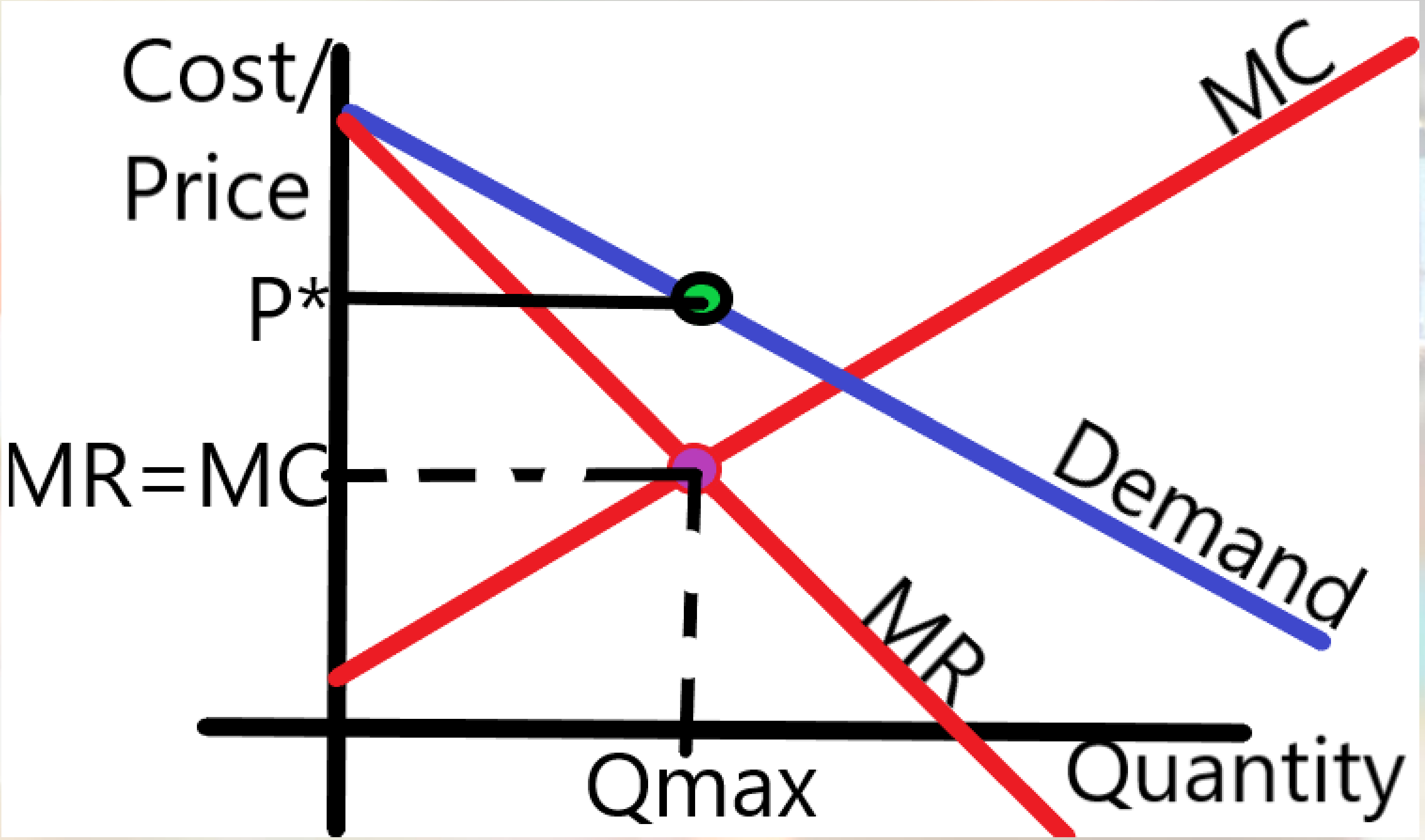
P>MR Ch15-2b

P>MC (at profit max) CH15-3a

Examples:

-Utilities, Patents

Monopoly Graph



Terms

ATC ➔ Average Total Cost

MC➔ Marginal Cost

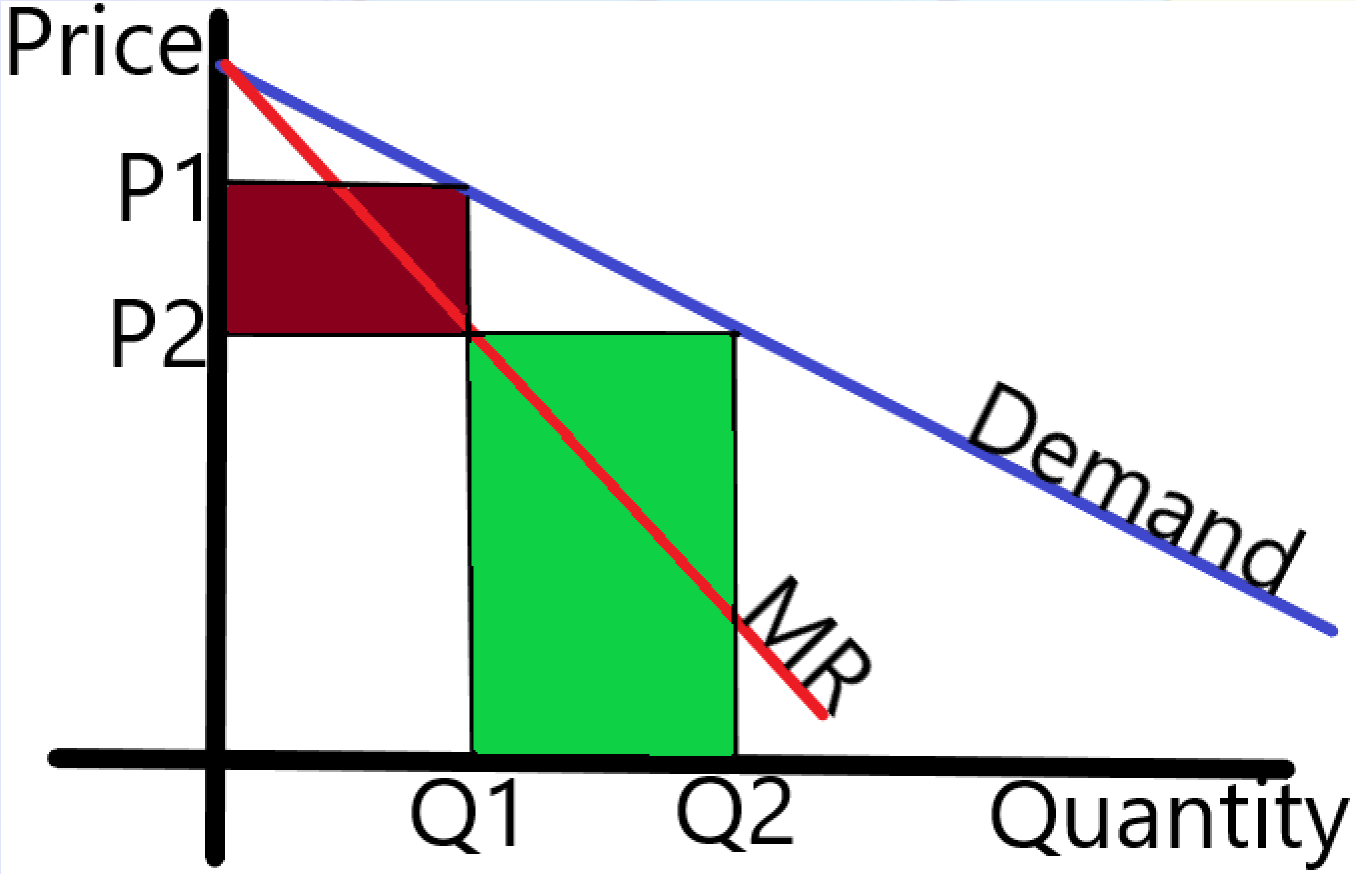
MR➔ Marginal Revenue

P*=Market Price

- Price charged by the monopoly

Qmax➔ profit maximizing Quantity

Why is Marginal Revenue less than Price?



When the Monopoly firm produces more units, the Price must fall (P1→P2)

The downward-sloping Demand curve IS the Market Demand curve

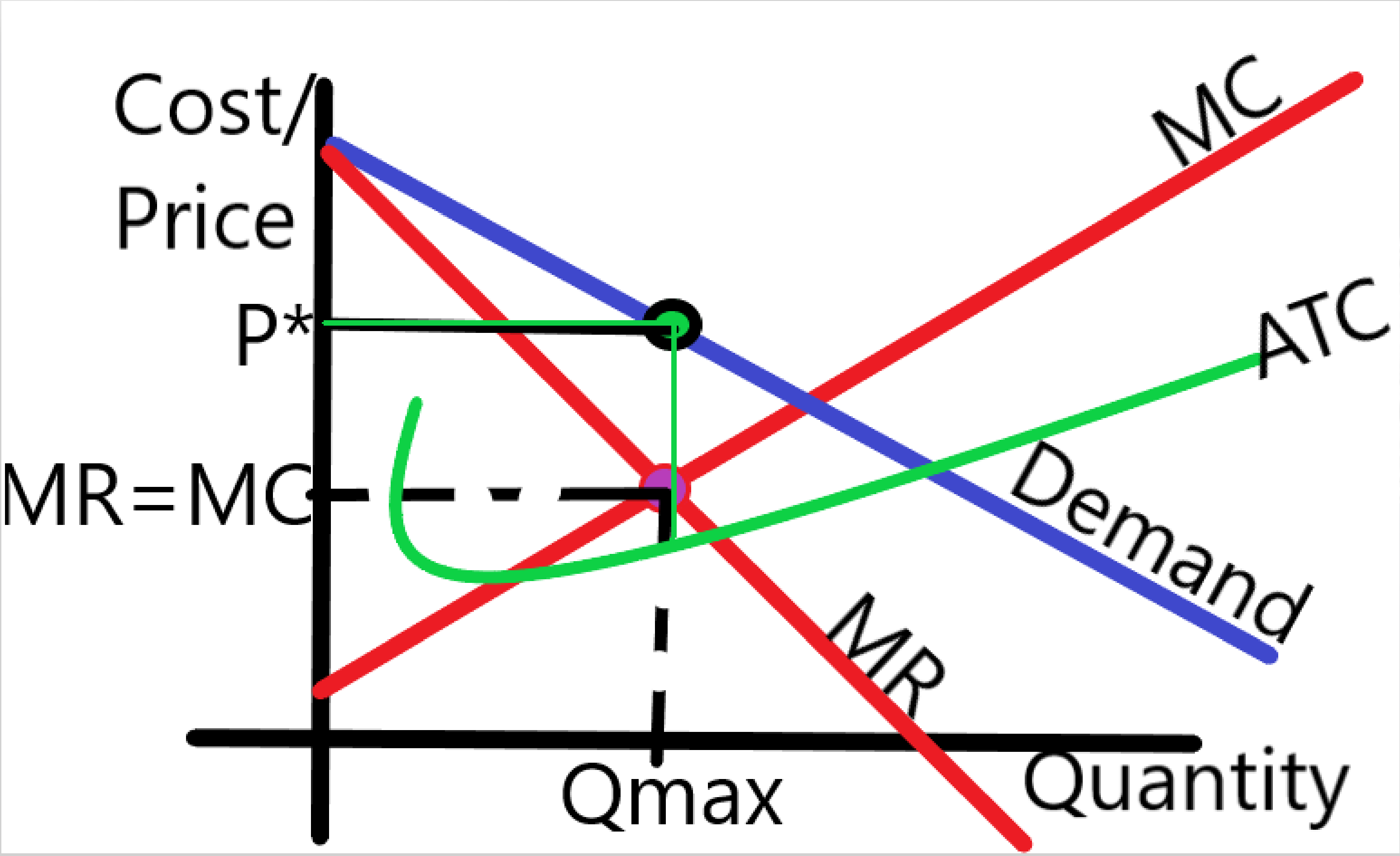
So greater Quantity lowers Price

The GREEN area is the gained revenue from selling more units

The RED area is the loss of revenue caused by the fall in price

The combined effect IS the Marginal Revenue curve being below Demand

Monopoly Graph



Terms

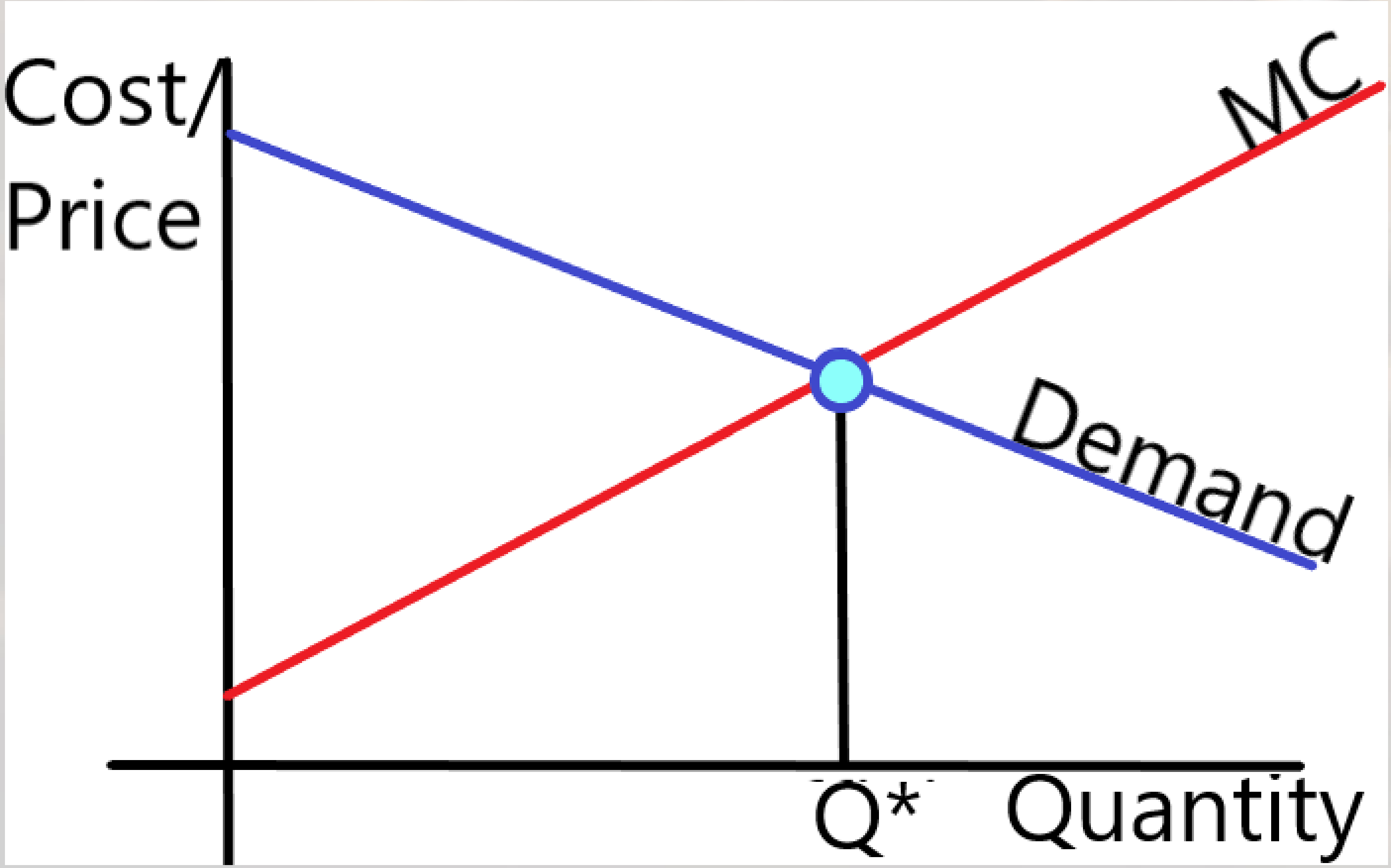
How much profit?

-Determined by P* minus Average Total Cost

No difference between short-run and long-run

-No competition can enter so profits remain

Social Cost of Monopoly



Terms

Q*=Efficient Quantity

-Best for society/customers

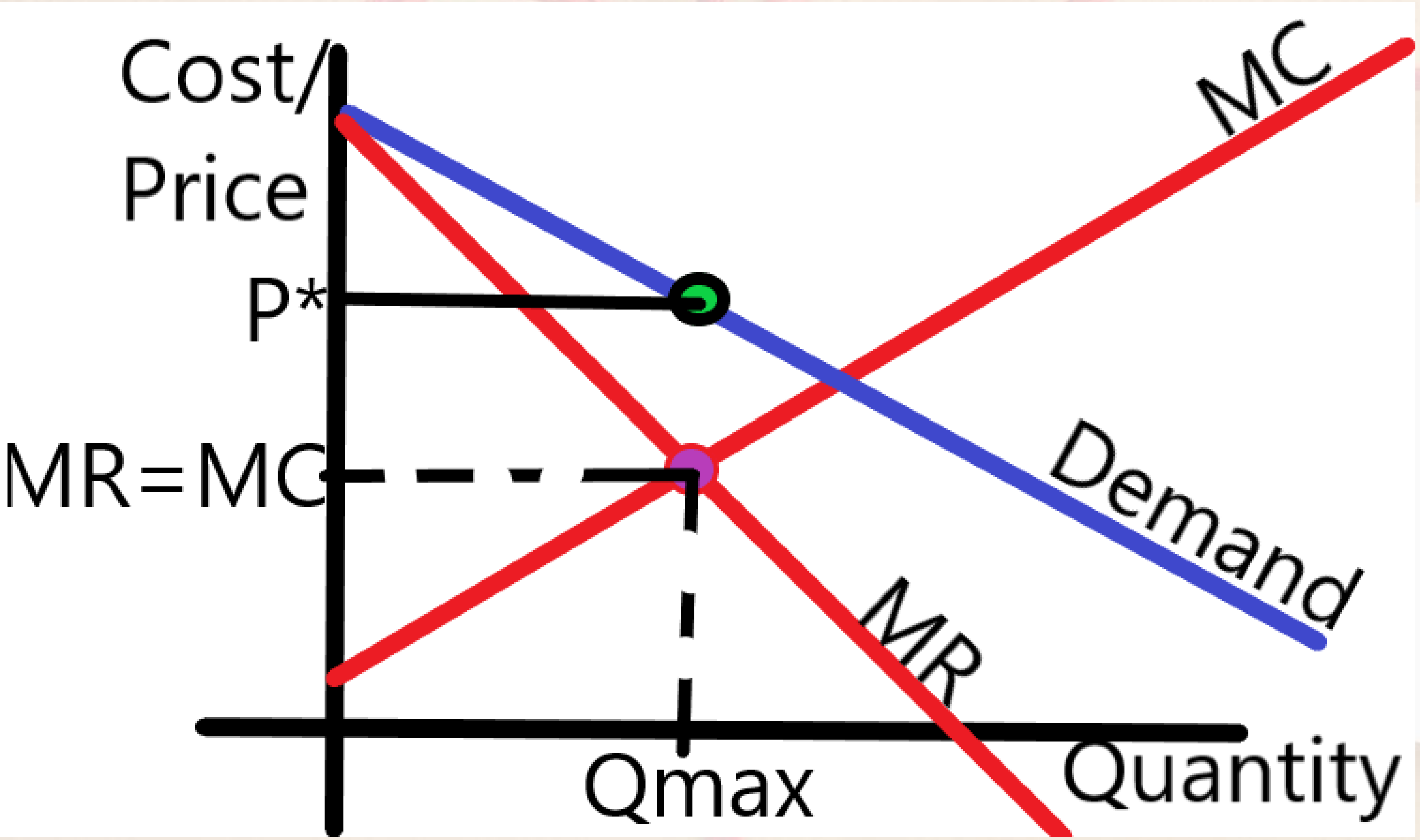
-higher quantity

MC=Marginal Cost

Get Market Supply by adding all MC curves

-With only one firm, their MC curve IS the market Supply curve

Barriers to Entry



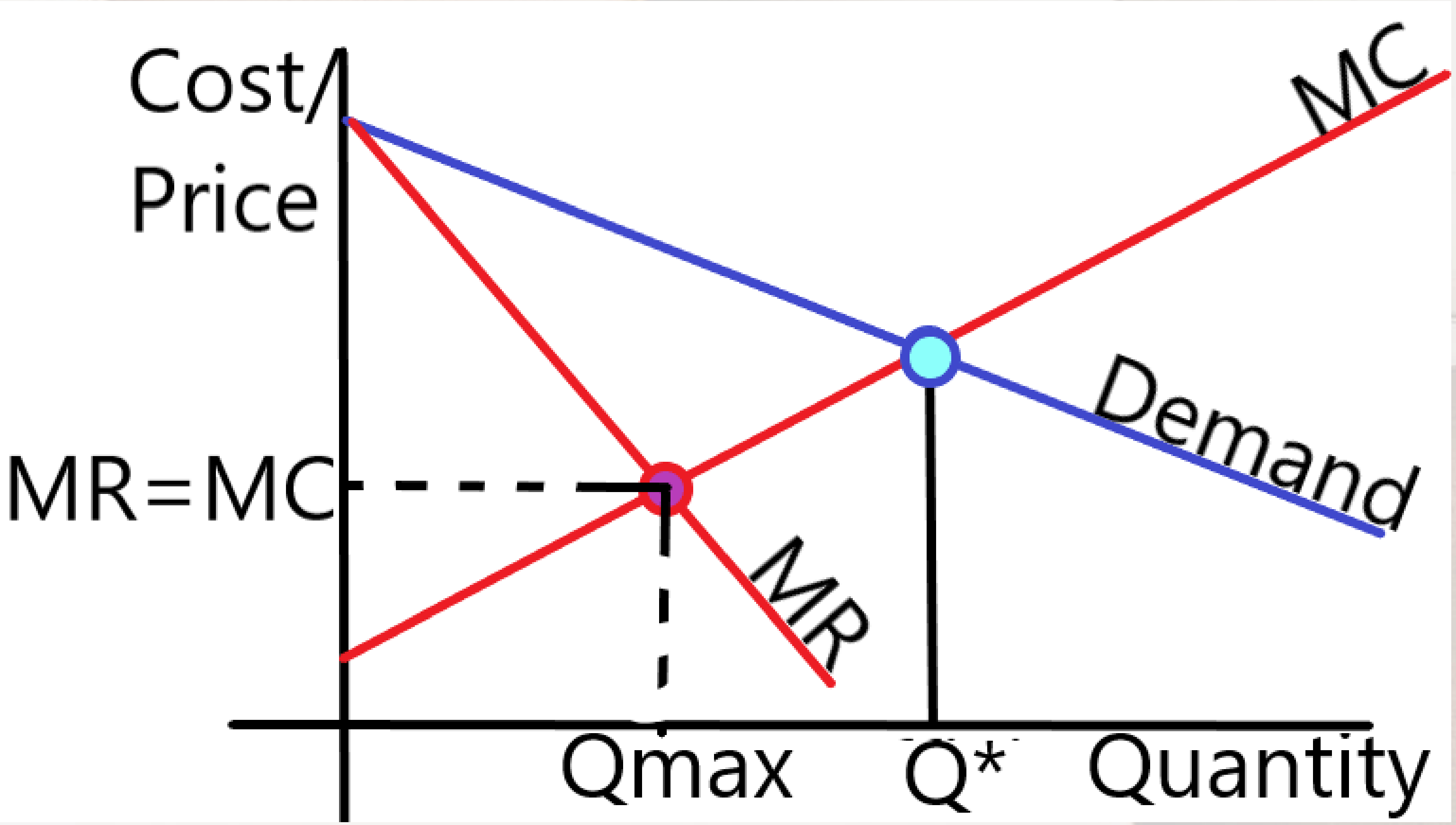
Causes of Monopoly Barriers

1) **Owning a Resource**
   - **Single owner of diamond mine**

2) **Government regulation**
   - **Patents, Copyrights, etc.**

3) **Natural monopoly**
   - **Also called a Production Process Monopoly**
   - **Caused by Economies of Scale**

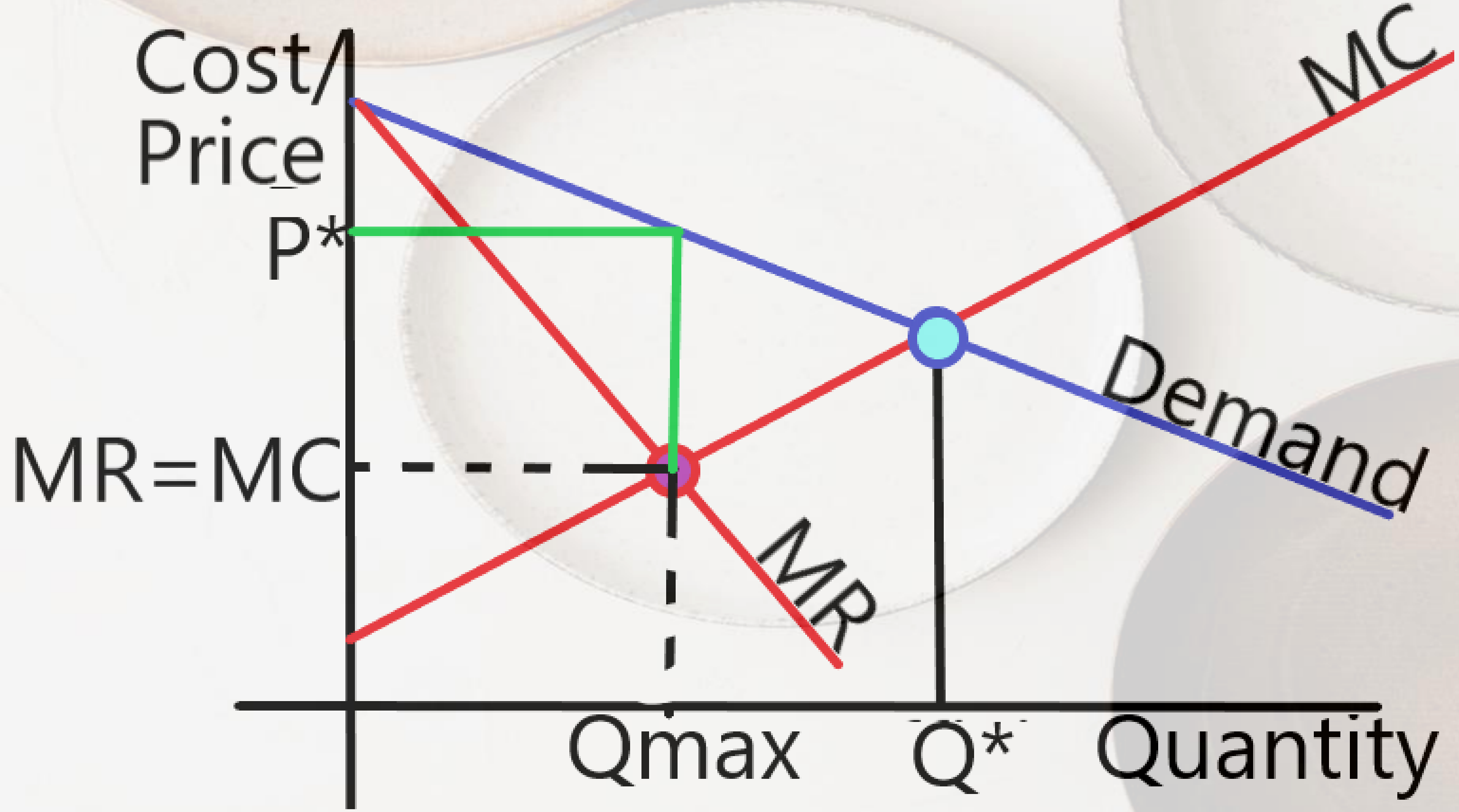
Social Cost of Monopoly



Profit max

Qmax➔ Lower quantity

Social Cost of Monopoly

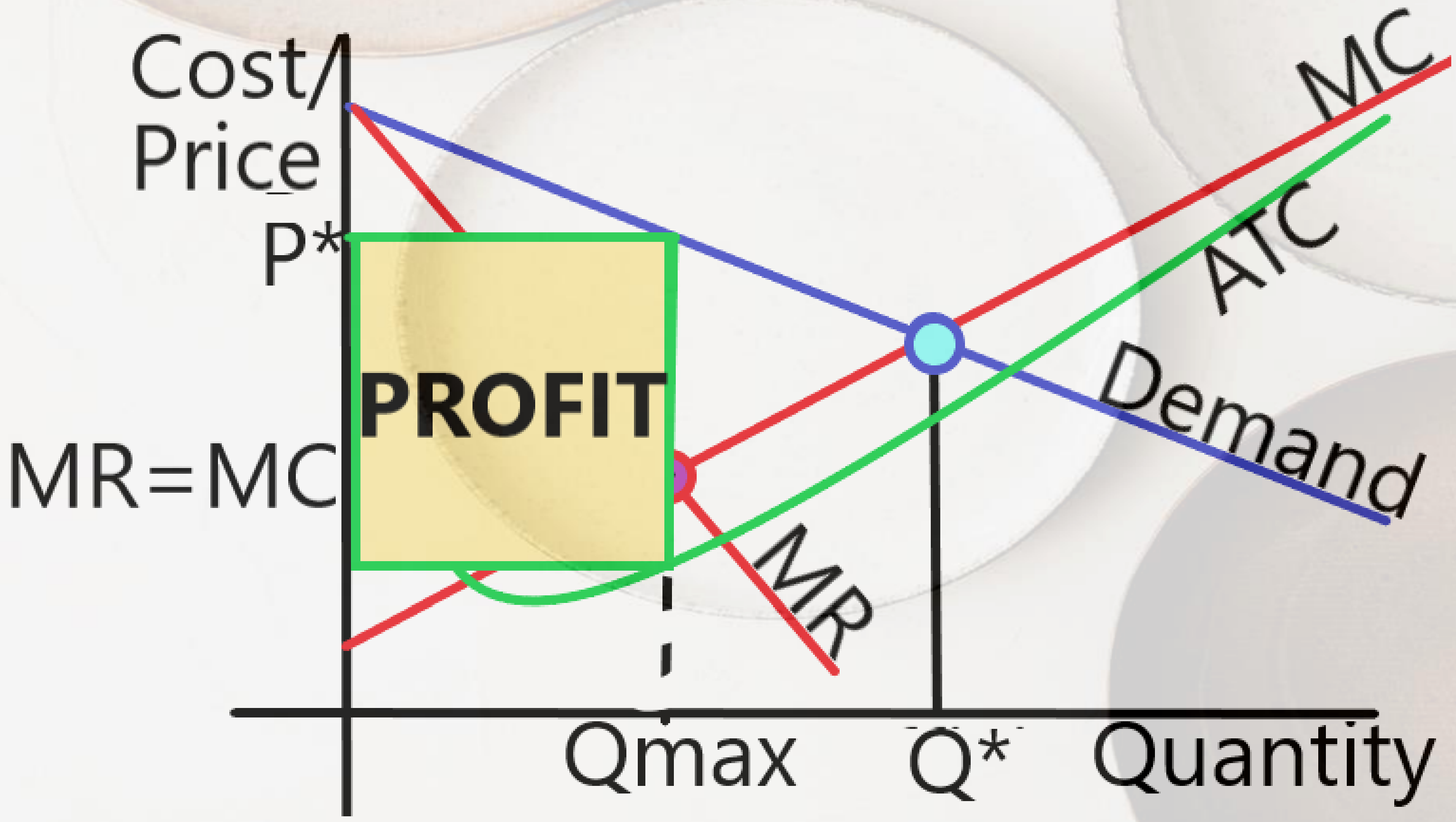


Profit max

Qmax➔ Lower quantity

P*➔ The price they can charge at Qmax

Social Cost of Monopoly



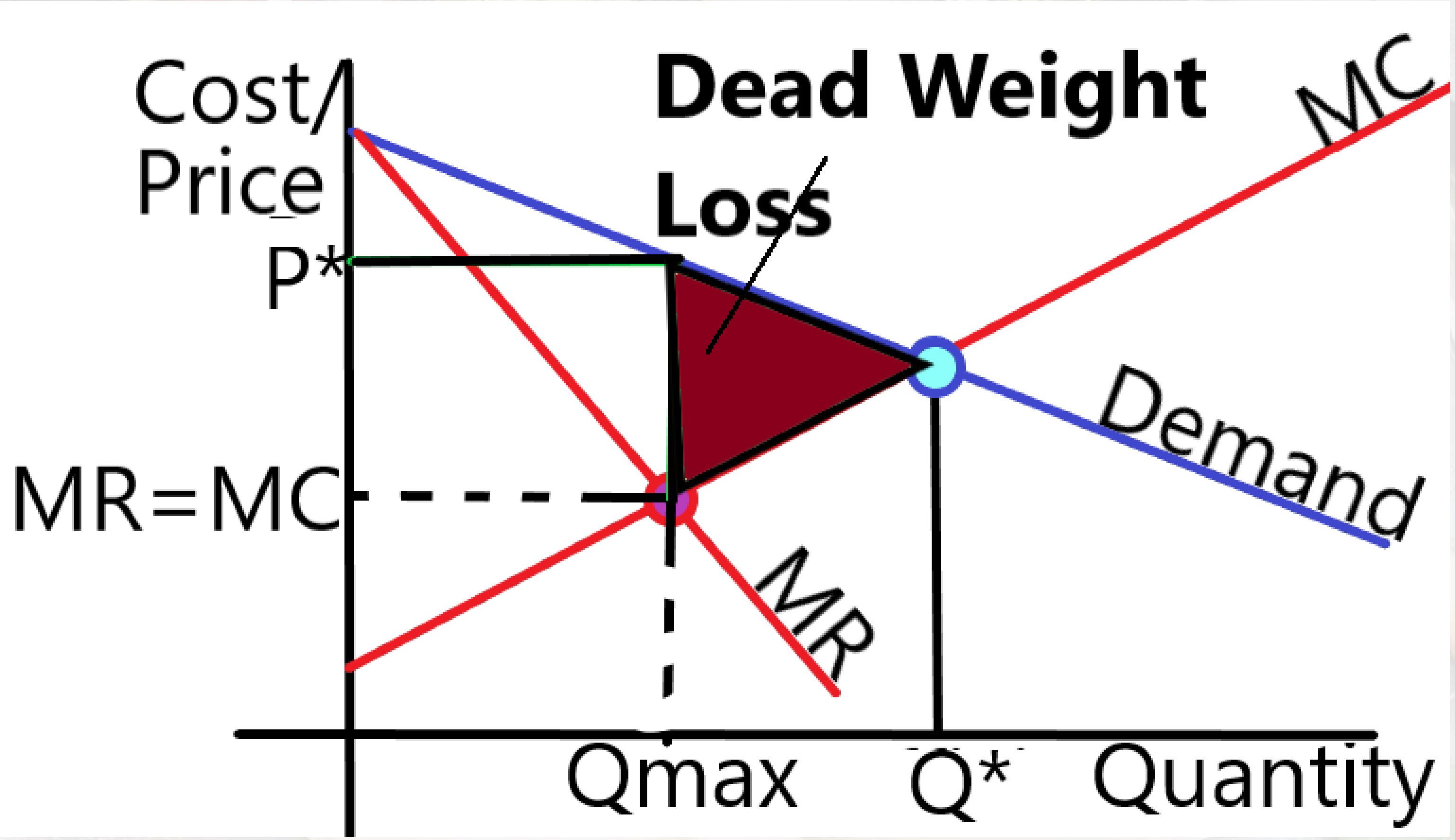
Profit max

Qmax➔ Lower quantity

P*➔ The price they can charge at Qmax

If we add in the Average Total Cost curve we can calculate profit

Social Cost of Monopoly



Profit max

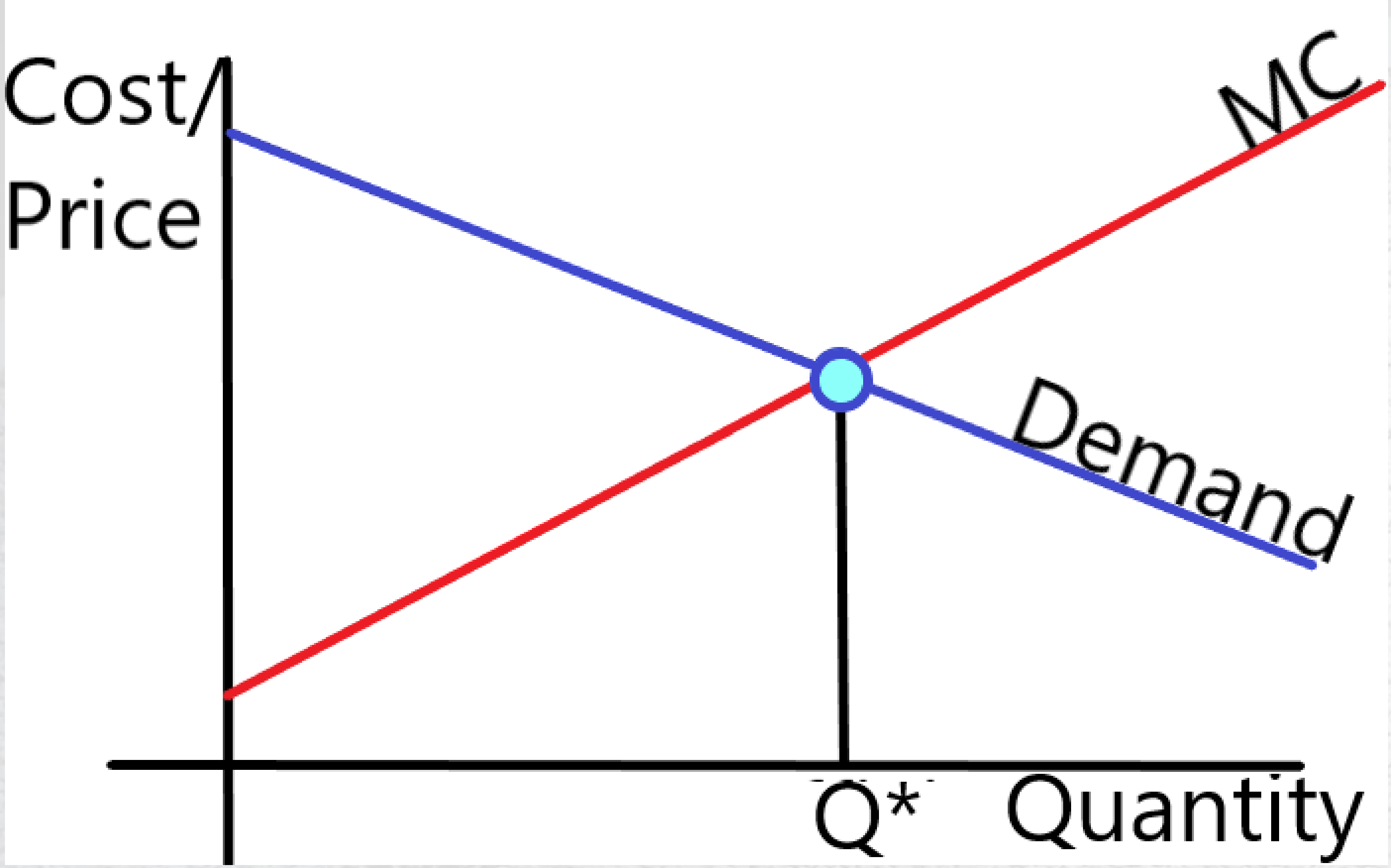
Qmax🡪 Lower quantity

P*🡪 The price they can charge at Qmax

Because the Monopoly reduces quantity

- Price rises
- Some customers are pushed out of the market

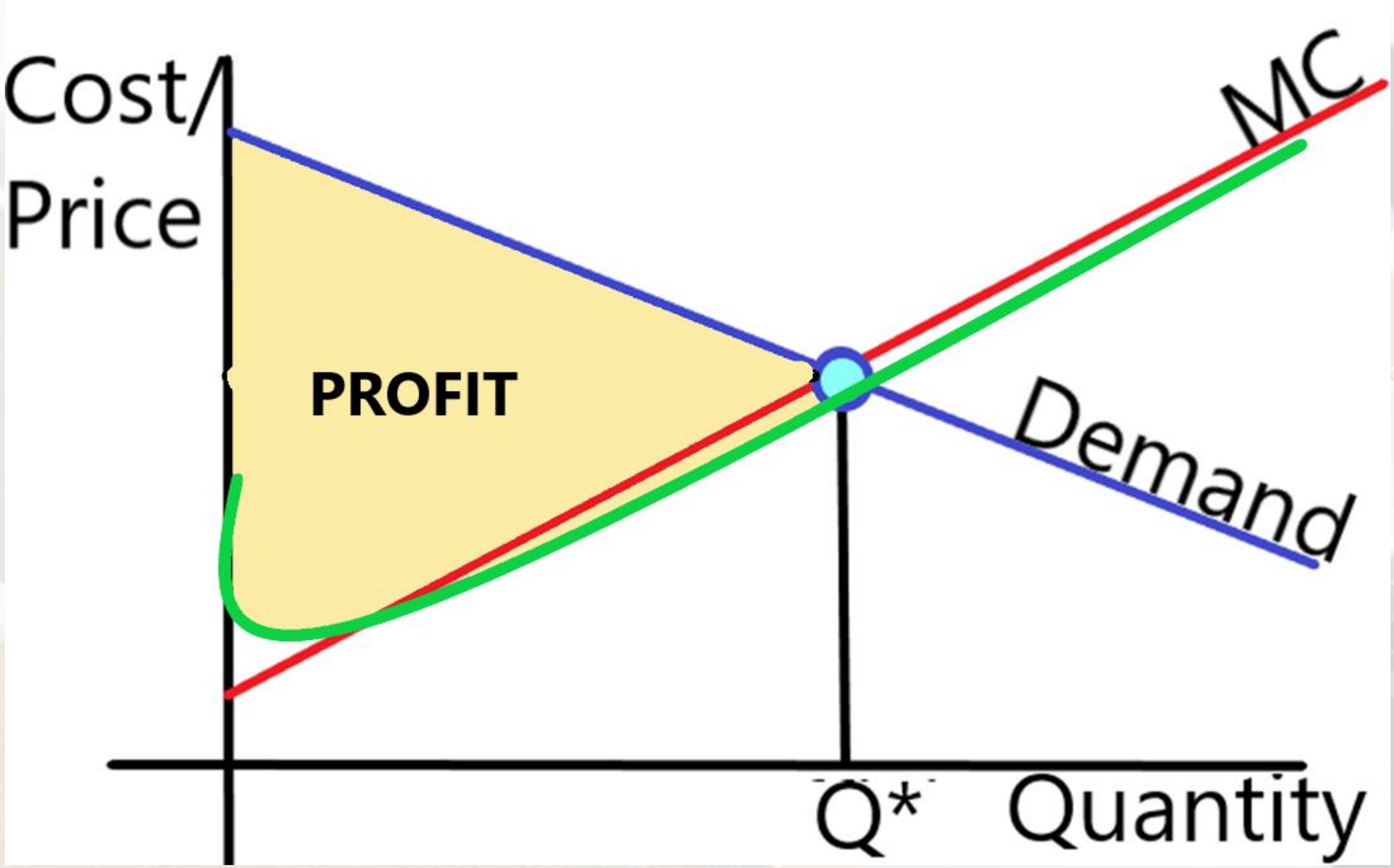
Social Cost of Monopoly



Two Ways to Correct Monopoly Inefficiency

1) Regulation
   - Force monopoly to produce Q* where MC=Demand
   - This will maximize Total Surplus
   - Would be chosen by a Benevolent Social Planner

Social Cost of Monopoly



Two Ways to Correct Monopoly Inefficiency

2) Price Discrimination

- **If there are No arbitrage opportunities**

- **Charges every customer their highest willingness to pay**

NO DEAD WEIGHT LOSS!!

CONCLUSION

Perfect Competition is Efficient Scale of production

Monopoly without regulation produces less than the Efficient Scale

Perfect Competition P=MR and so P=MC at profit max

Monopoly P>MR and so P>MC at profit max

Perfect Competition has no barriers

Monopoly has impossible barriers